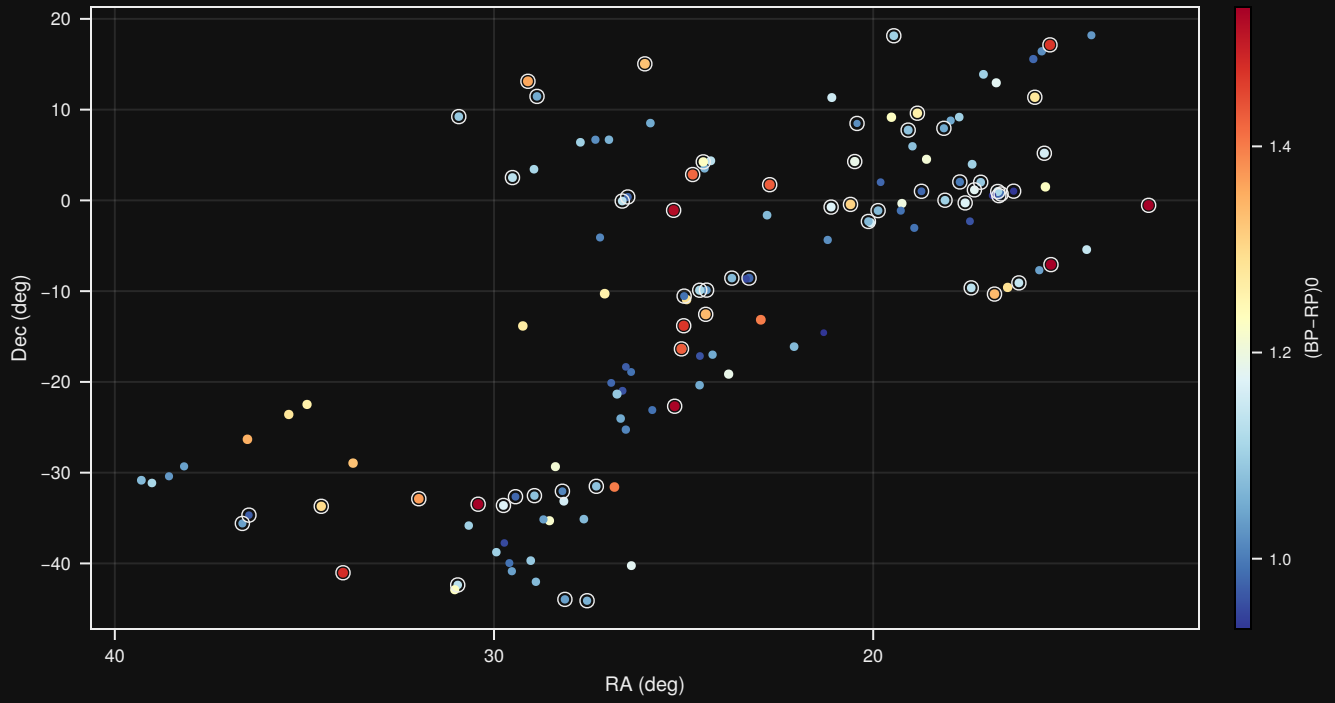
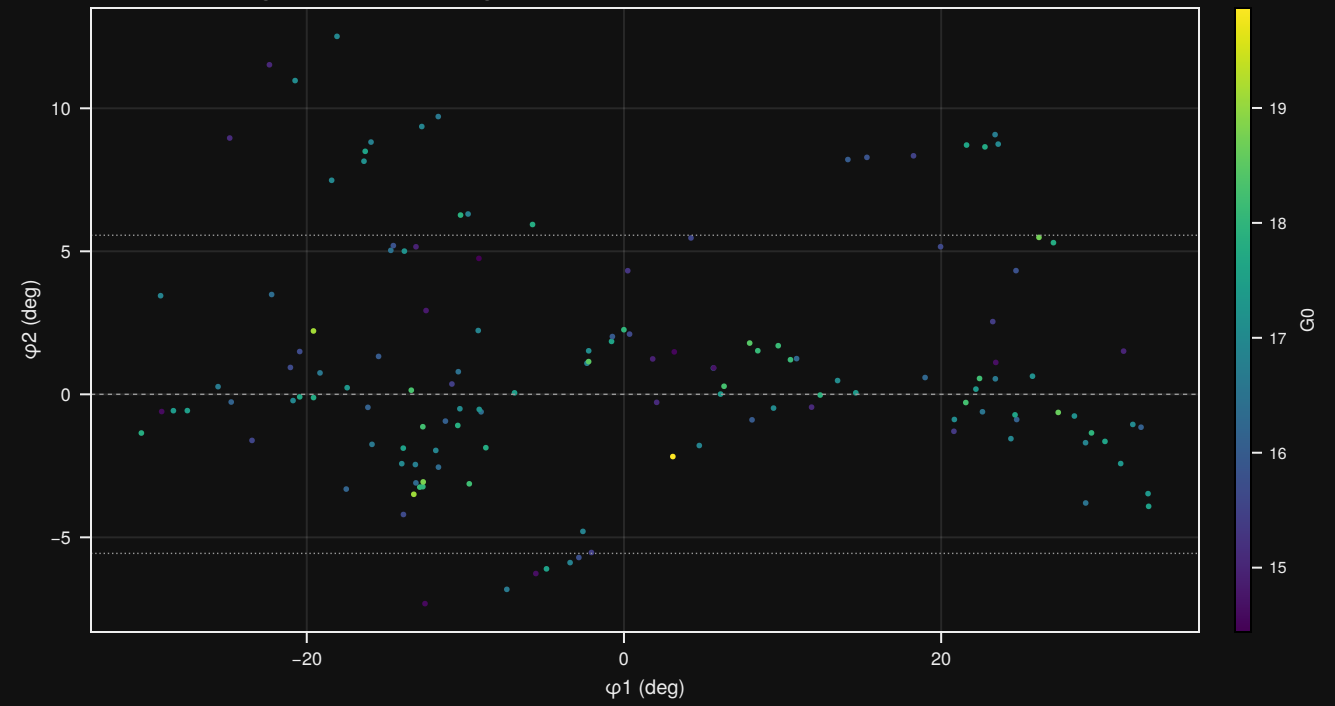
Kwando

MEMBERSHIP	
Stars in catalog	138
with Gaia GSP-Phot	114 (83%)
with radial velocity	62
Stream id (sID)	6
GEOMETRY · great-circle fit	
Centroid RA, Dec	24.2, -8.6 deg
Centroid l, b	168.0, -67.7 deg
Pole RA, Dec	110.1, 13.2 deg
Length (φ1, 2–98%)	60.8 deg
Width σ(φ2), robust	2.78 deg
KINEMATICS	
pmRA* med ± σ	0.80 ± 0.12 mas/yr
pmDE med ± σ	-0.45 ± 0.14 mas/yr
μφ1* med ± σ	0.63 ± 0.13 mas/yr
μφ2 med ± σ	0.65 ± 0.11 mas/yr
Parallax (cat.) med	0.016 ± 0.059 mas
Parallax (wtd, ZP-corr.)	0.034 ± 0.011 mas
Distance	d = 29.0 ± 9.4 kpc
Physical length	30.80 kpc
RV median ± σ(MAD)	-96.9 ± 41.3 km/s
RV gradient dV/dφ1	5.19 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	14.4–19.9 (16.9)
(BP−RP)0 median	1.10
GSP-Phot [M/H] (n=114)	-1.51 ± 1.37
GSP-Phot Teff median	5093 K
RUWE > 1.4	0.0 %
PM clusters (largest)	9 (19 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	138 | 138
n_RV catalog | paper	62 | 62
<[Fe/H]> paper	-2.29
Stellar mass (paper)	40000 Msun
Pole offset vs paper	3.8 deg
Remark	Very wide, rough [Fe/H] estima

Sky map · catalog members (n = 138) · size ∝ brightness · ring = has RV

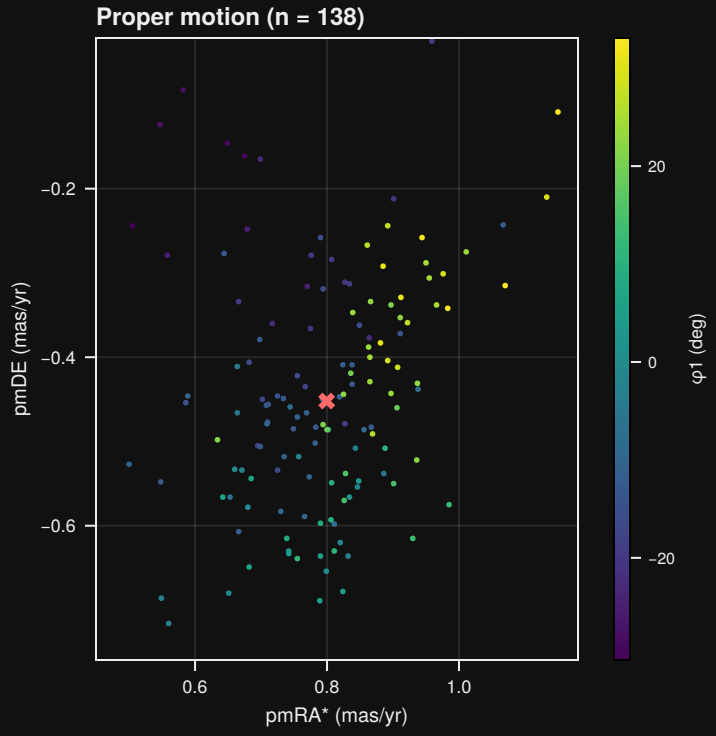


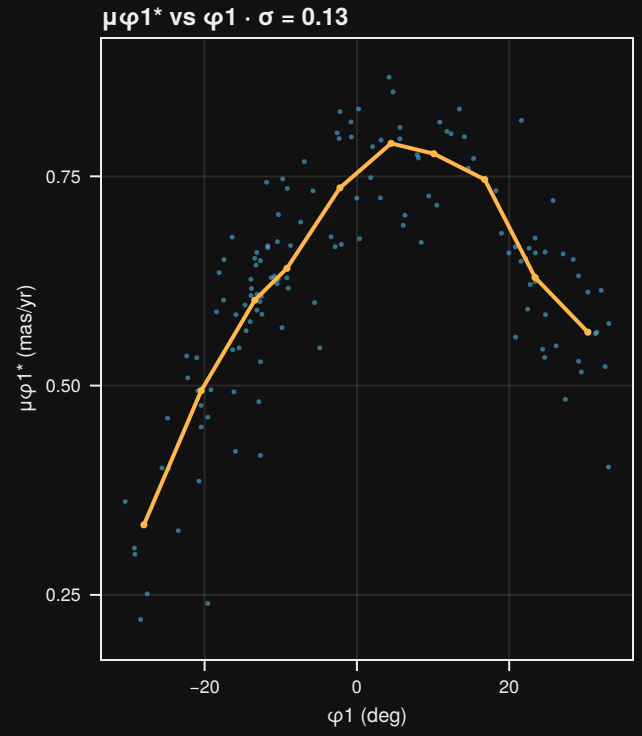
Stream frame (great-circle fit) · length 60.8° · width σ 2.78°

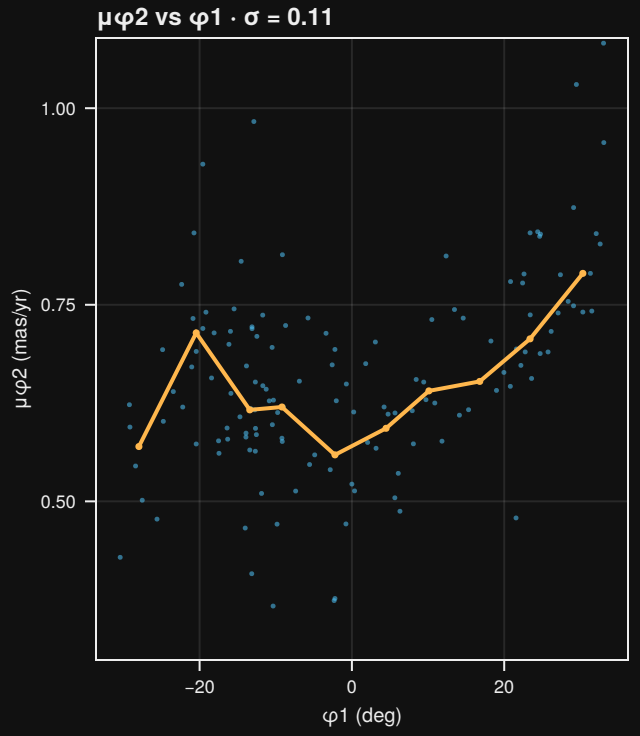


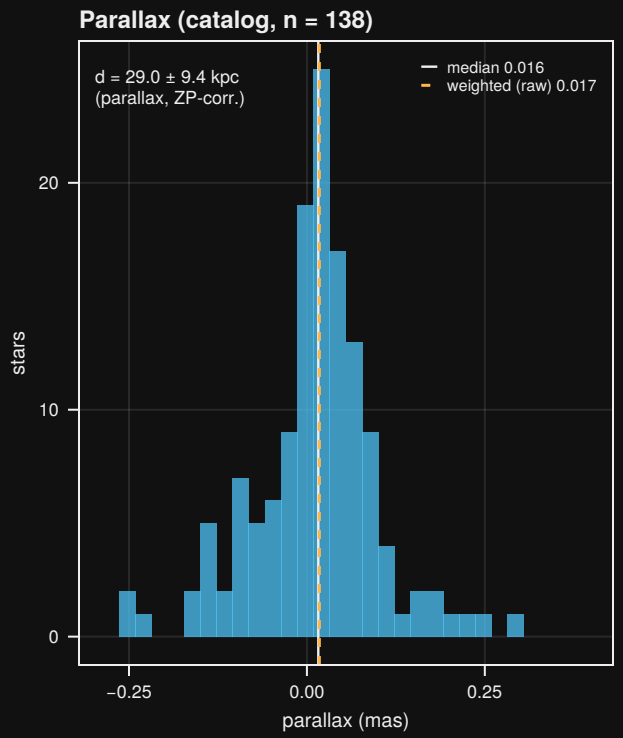
Kwando

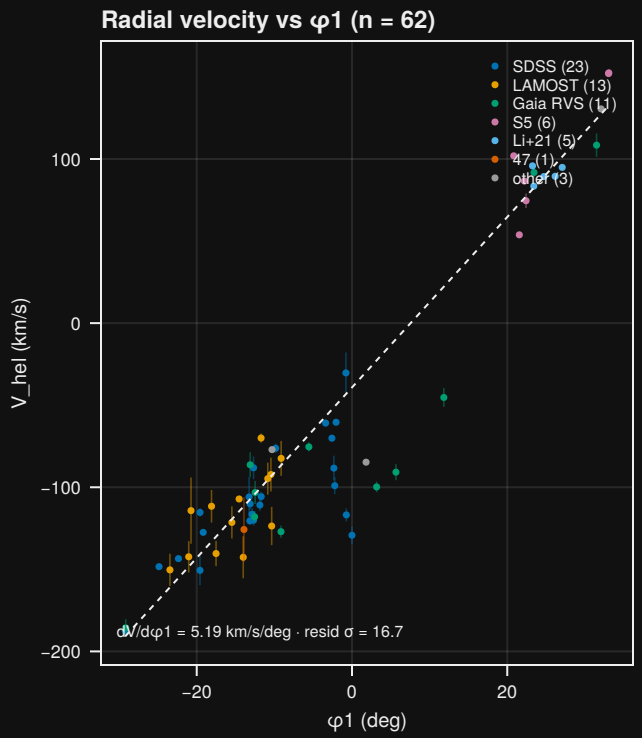
Kinematics · proper motion · parallax · radial velocity

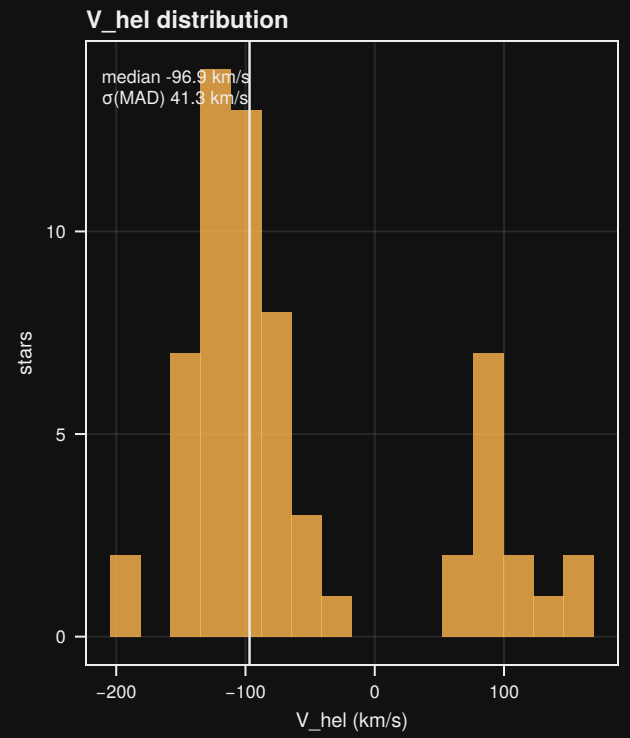






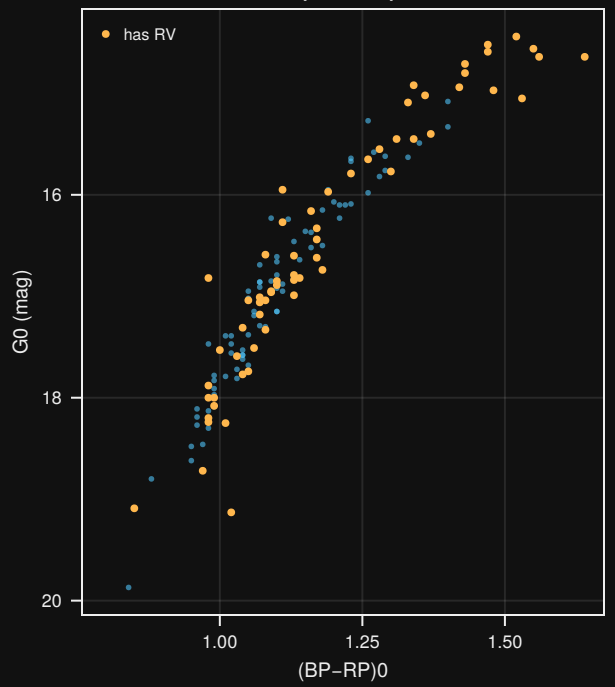




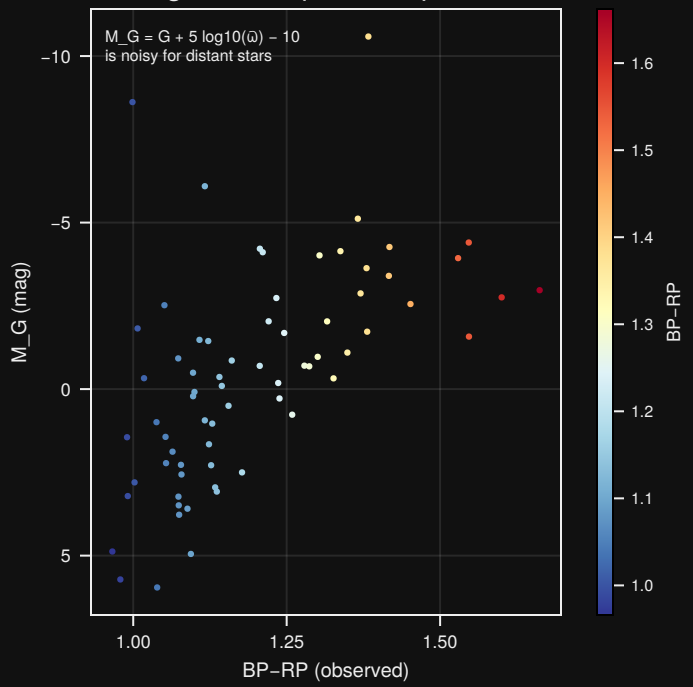


Kwando

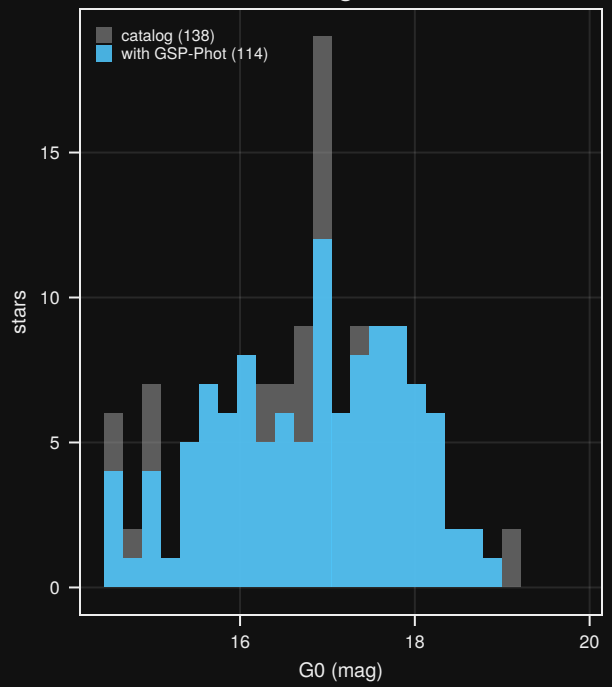
CMD · dereddened (n = 138)



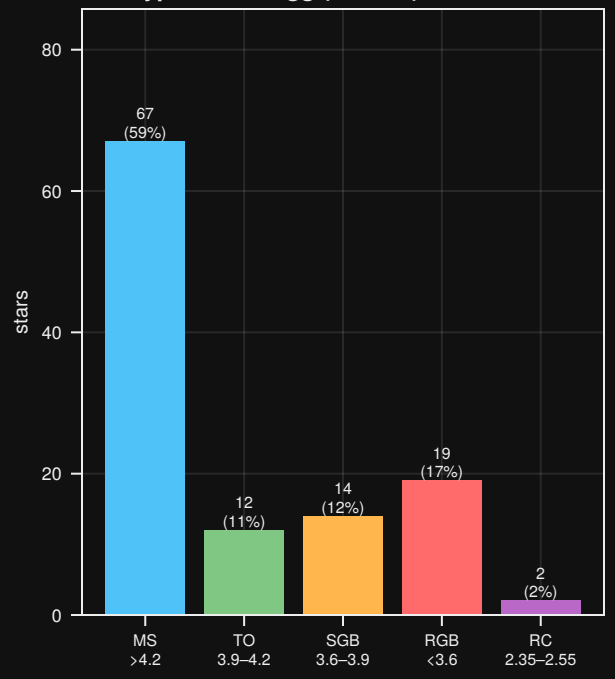
HR diagram · ϖ>0 (n = 69/114)



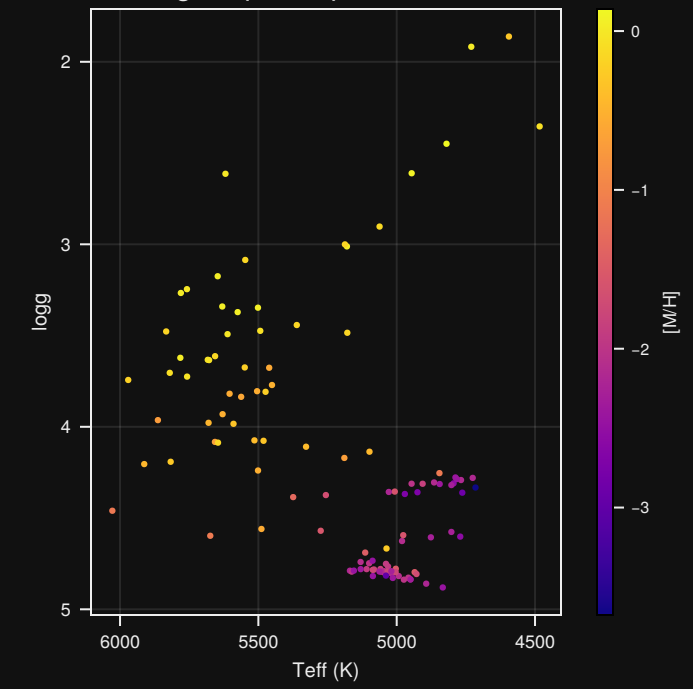
G0 & GSP-Phot coverage



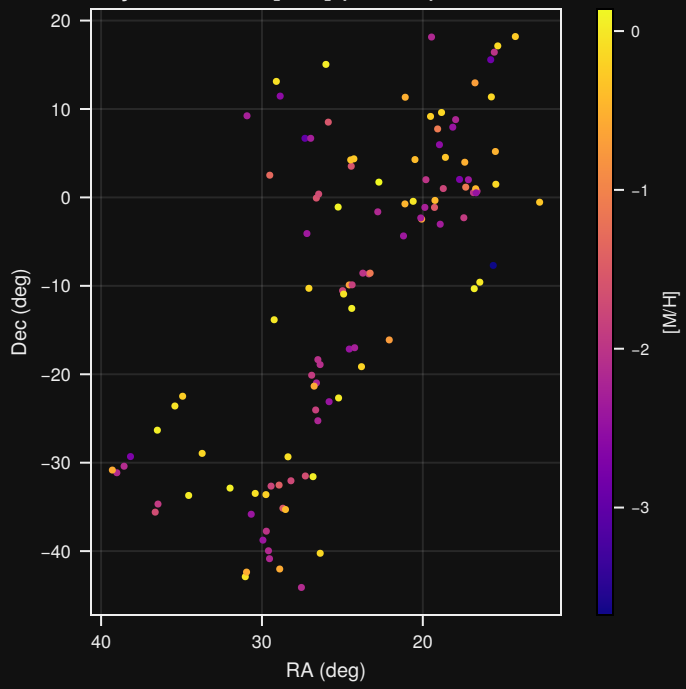
Star types from logg (n = 114)



Kiel diagram (n = 114)



Sky · GSP-Phot [M/H] (n = 114)

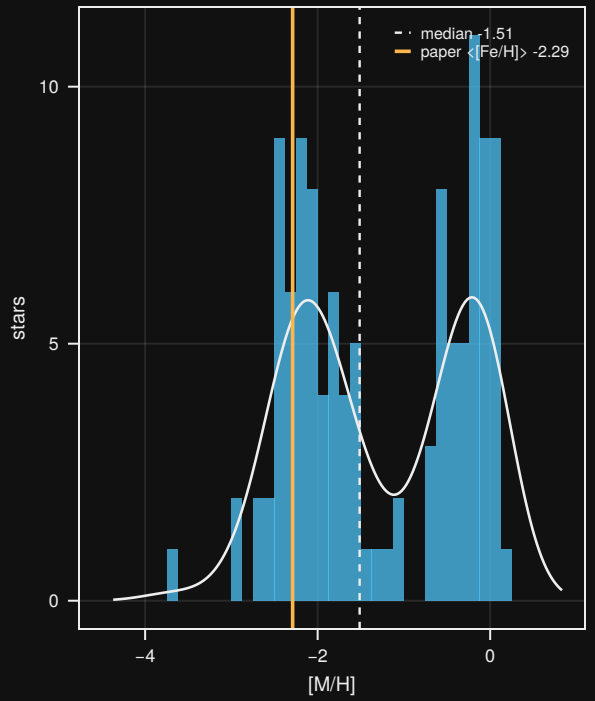


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

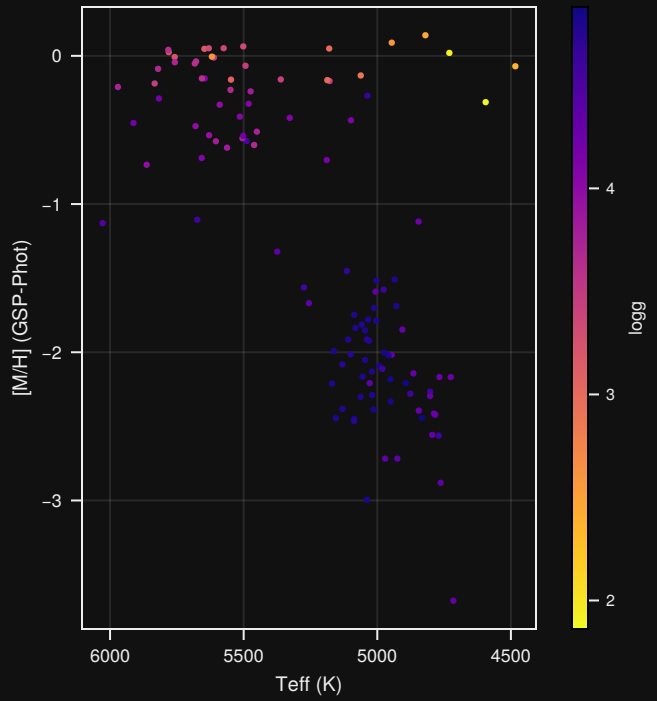
Kwando

Metallicity · age · astrometric quality

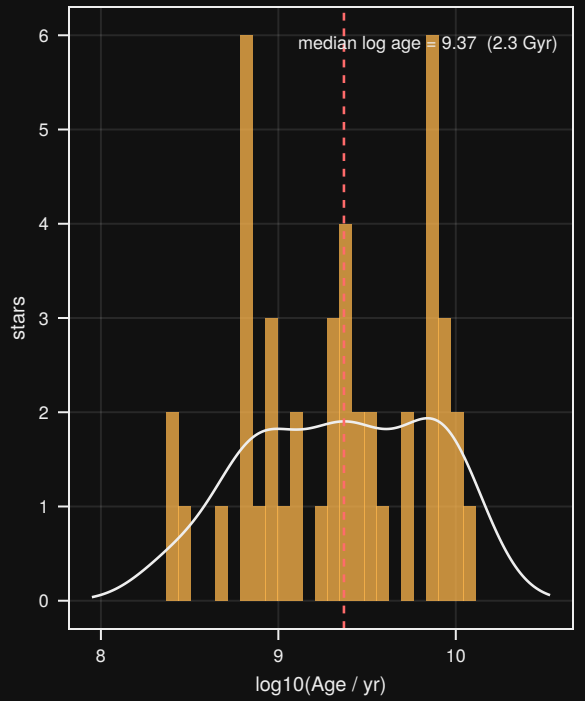
[M/H] GSP-Phot (n = 114)



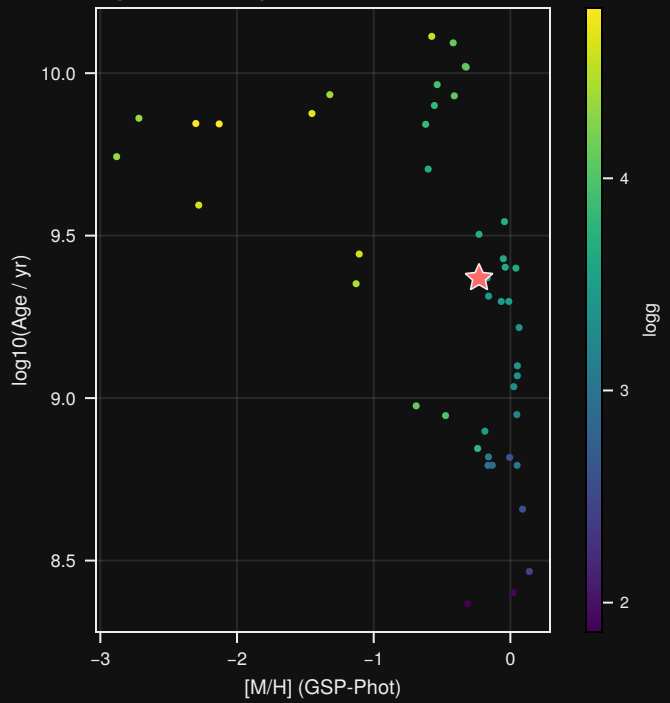
Teff vs [M/H] (n = 114)



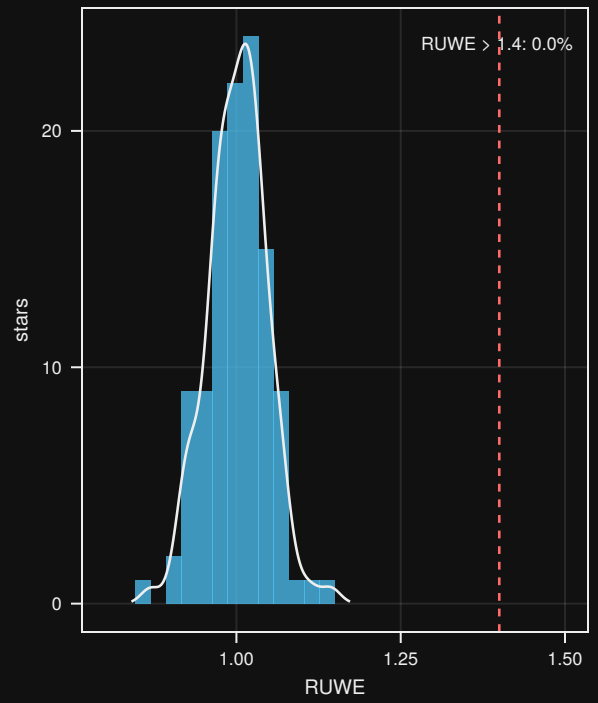
FLAME age (n = 45)



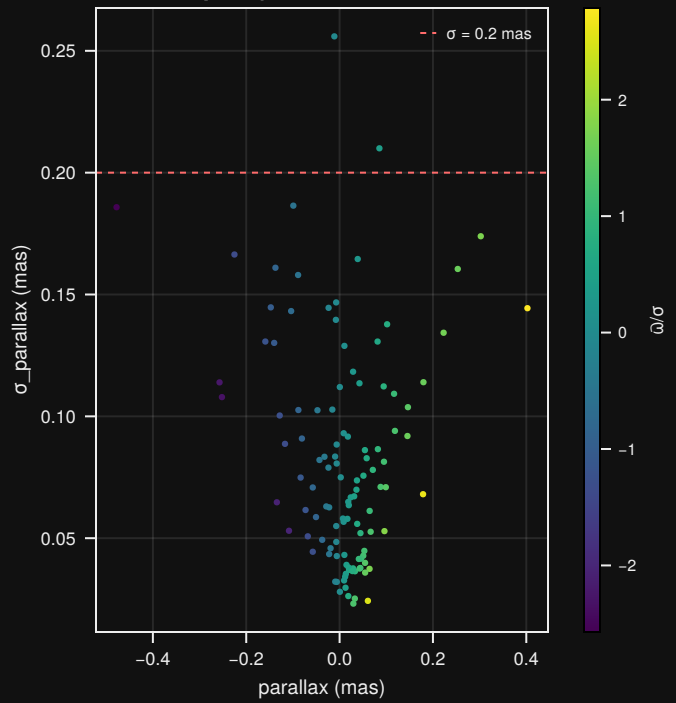
Age–metallicity (n = 45)



RUWE (n = 114)

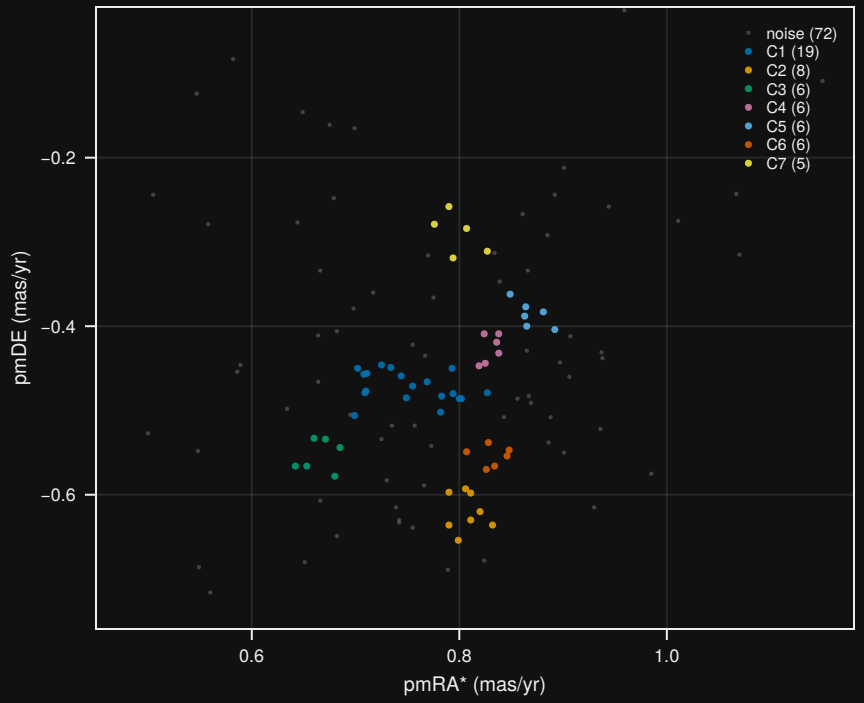


Parallax quality (n = 114)

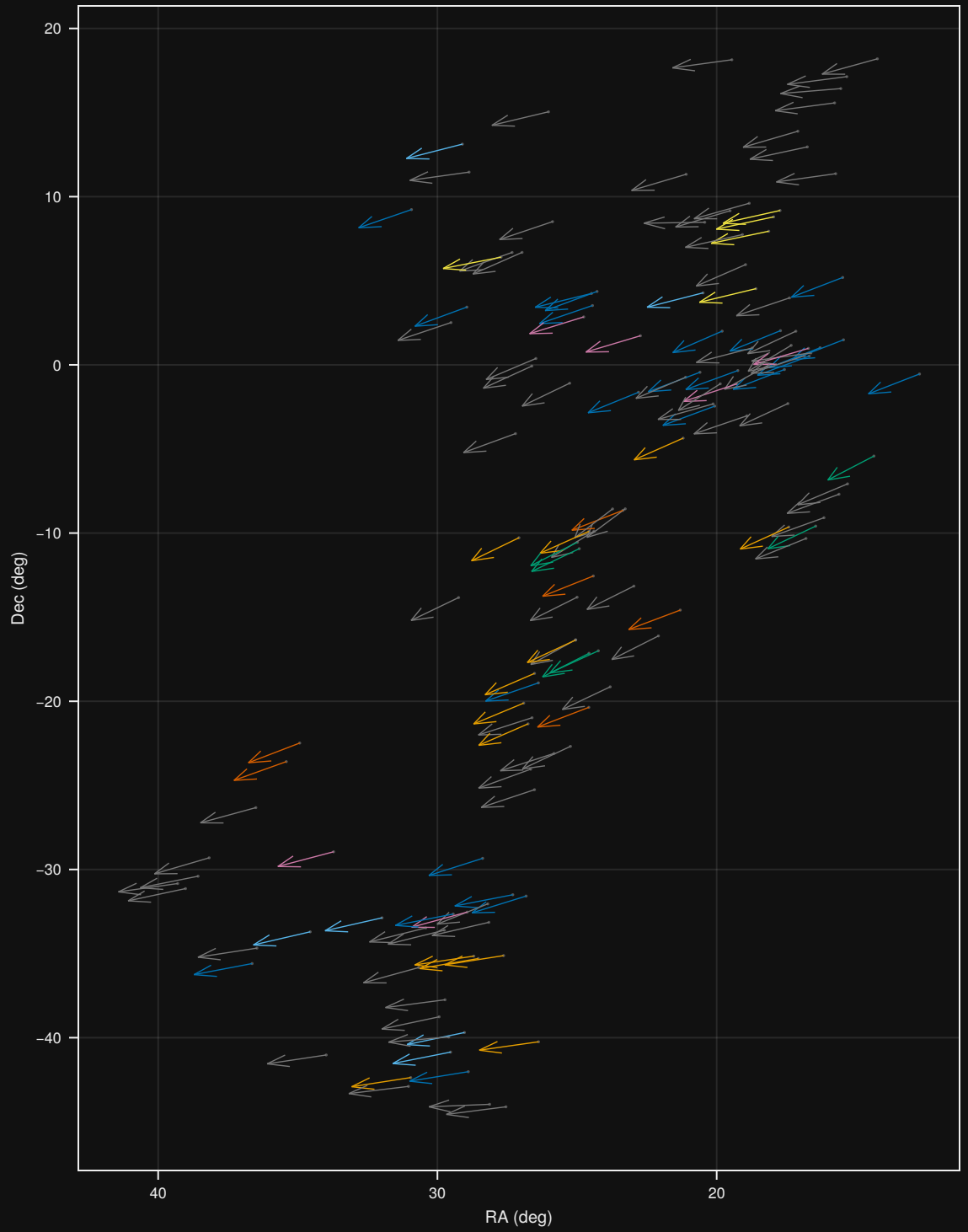


Kwando

PM clustering · HDBSCAN · 9 cluster(s) · noise 52%



Proper-motion directions (arrows) coloured by cluster



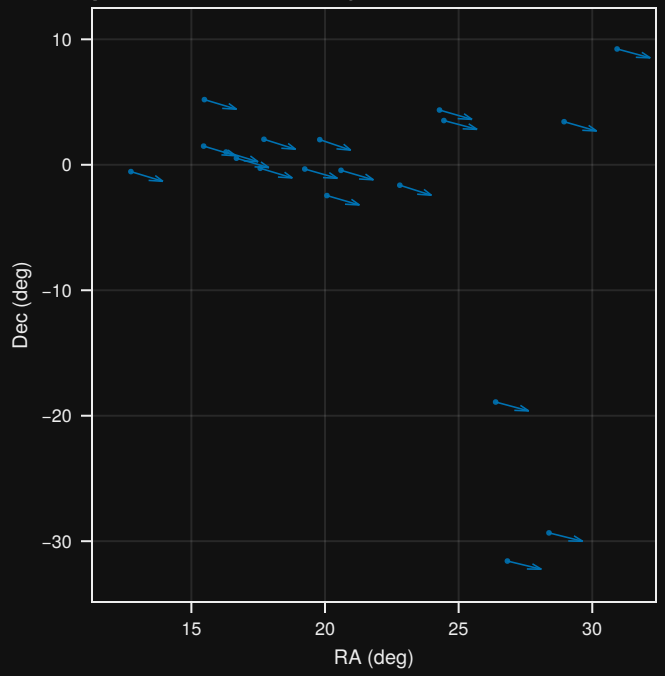
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	72	0.77	-0.44	0.15	-1.85	35
C1	19	0.75	-0.48	0.06	-0.57	8
C2	8	0.81	-0.62	0.02	-1.30	3
C3	6	0.67	-0.55	0.02	-1.81	1
C4	6	0.83	-0.43	0.01	-0.45	4
C5	6	0.86	-0.39	0.01	-0.07	4
C6	6	0.83	-0.55	0.01	-0.14	1
C7	5	0.79	-0.28	0.02	-2.21	1
C8	5	0.91	-0.35	0.00	-1.67	4

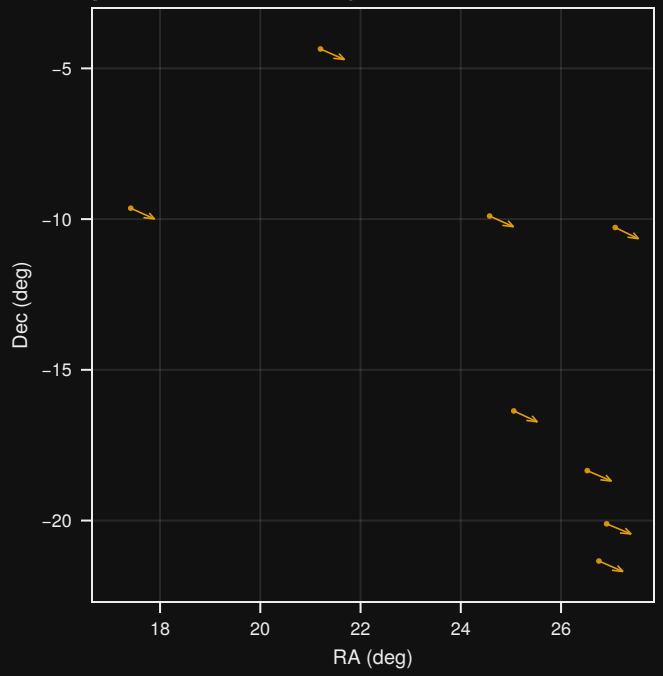
Kwando

Per-cluster proper-motion vector maps

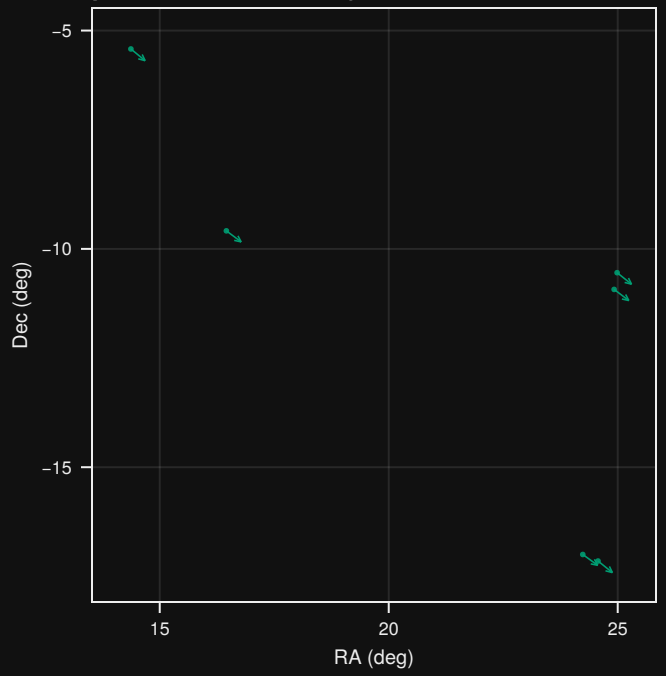
Cluster C1 · n = 19
pm = (0.75, -0.48) mas/yr



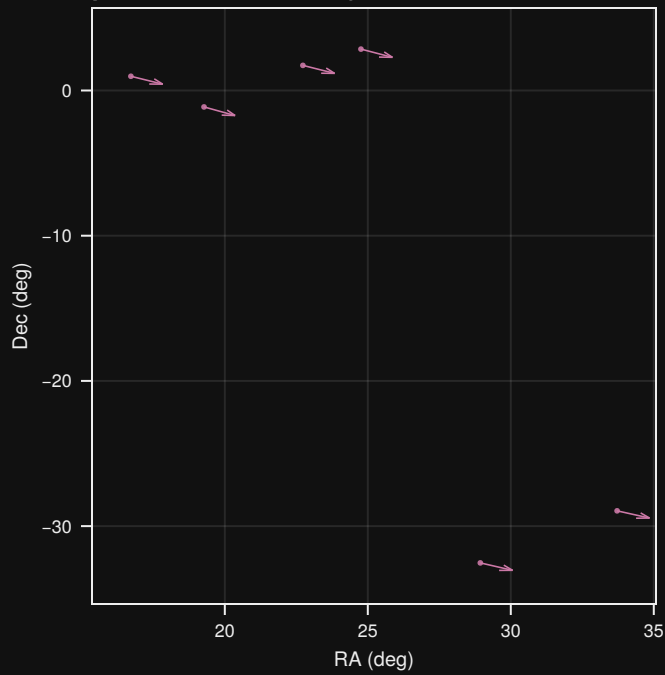
Cluster C2 · n = 8
pm = (0.81, -0.62) mas/yr



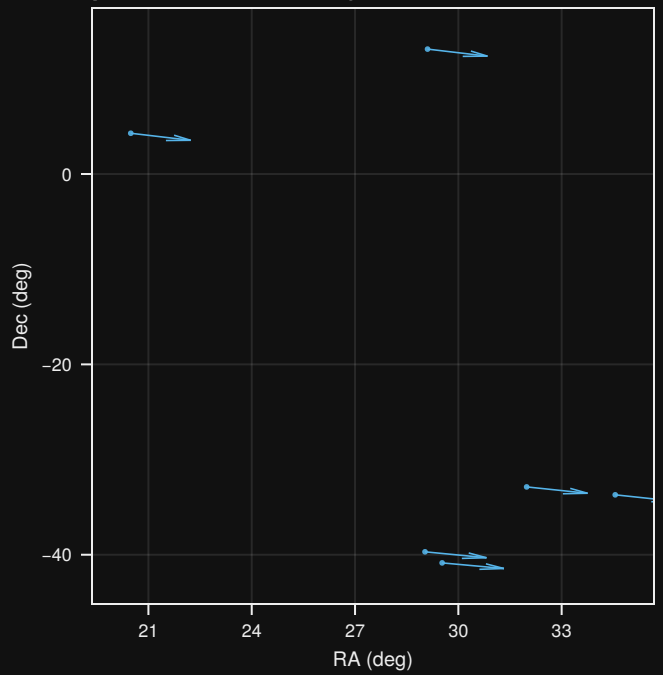
Cluster C3 · n = 6
pm = (0.67, -0.55) mas/yr



Cluster C4 · n = 6
pm = (0.83, -0.43) mas/yr



Cluster C5 · n = 6
pm = (0.86, -0.39) mas/yr



Noise · n = 72

